Mid Term: K-Means Clustering

TalentWorks Employee Segmentation

MBAI 5310G-001: AI Programming | Ontario Tech University

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1. Business Context

TalentWorks is an HR consultancy helping mid-size organisations reduce employee attrition. Rather than reacting to resignations after the fact, TalentWorks wants to proactively segment its workforce into meaningful groups so that targeted retention and engagement strategies can be applied to each segment. This report applies K-Means clustering to identify natural groupings of employees based on workplace characteristics. Clustering is unsupervised: the *Left_Company* column is excluded from training and used only after clustering to validate whether the segments are meaningful for attrition risk.

2. Dataset

Property	Detail
File	talentworks_employee_attrition_dataset.xlsx
Records	286 employee profiles
Columns	23 (12 numerical selected for clustering)
Target (validation only)	Left_Company (Yes / No)
Attrition rate	91 of 286 employees left (31.8%)

3. Methodology

Feature Selection: 12 numerical features selected: Age, Monthly Income, Tenure, Distance from Home, Commute Time, Job Satisfaction, Work-Life Balance, Training Hours, Absences, Performance Rating, Manager Feedback Score, and Engagement Score.

Categorical Features Excluded: Department, Job Role, Education Level, Employment Type, Overtime, Remote Work Available, Promotion Last 2 Years, Marital Status, and Workload Level were excluded. K-Means requires numerical input, and one-hot encoding these columns would have expanded the feature space significantly, making clusters harder to interpret. The trade-off is simplicity and interpretability at the cost of some HR context.

Missing Value Handling: Two columns (Monthly Income and Manager Feedback Score) contained missing values, imputed using column medians.

Feature Scaling: StandardScaler applied to bring all features to mean 0, standard deviation 1, preventing high-range features such as Monthly Income from dominating distance calculations.

Cluster Count Selection: Both the Elbow Method and Silhouette Score evaluated k = 2 to 8. Both metrics peaked at k = 2 and k = 3 (silhouette = 0.1120). k = 3 was selected because it provides more granular, actionable HR segments. Two clusters would produce groupings too broad for targeted retention strategy.

Model: KMeans(n_clusters=3, random_state=42, n_init=10). Segments validated post-hoc using attrition rates.

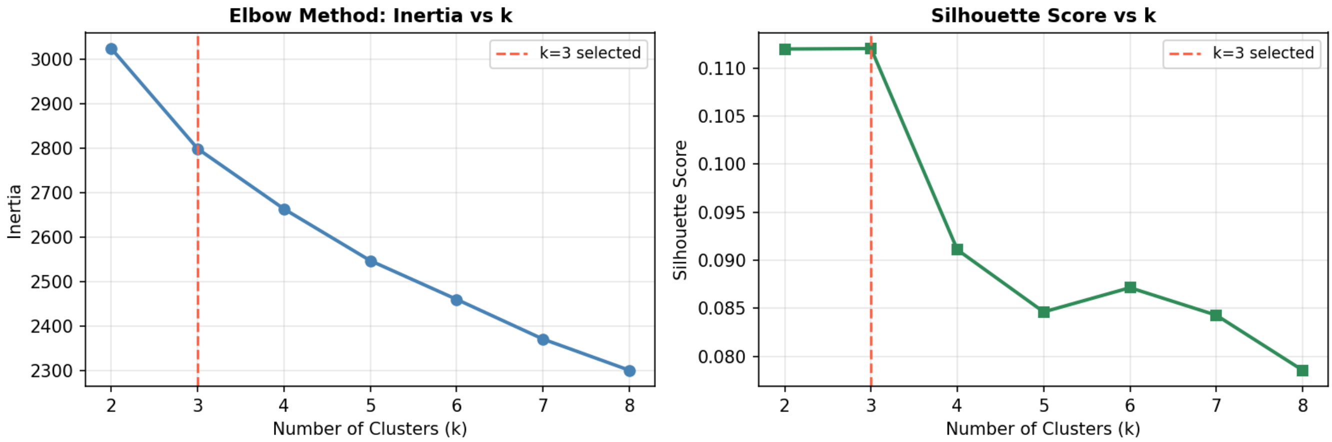

Figure 1. Elbow Method and Silhouette Score confirming k = 3.

4. Cluster Profiles

Feature	High-Performing Engaged	High-Commute At-Risk	Senior Long-Tenured
Age	36.38	35.62	34.92
Monthly Income	4709.79	4930.13	6608.23
Tenure Months	22.86	22.16	83.72
Distance From Home km	8.40	26.38	15.15
Commute Time Minutes	26.83	69.61	43.78
Job Satisfaction	3.55	3.09	3.36
Work Life Balance	2.94	3.27	3.52
Training Hours Last Year	24.78	19.22	17.18
Absences Last Quarter	2.88	2.18	3.48
Performance Rating	3.73	3.13	3.68
Manager Feedback Score	3.55	3.16	3.14
Engagement Score	66.70	65.52	66.30

5. Visualisations

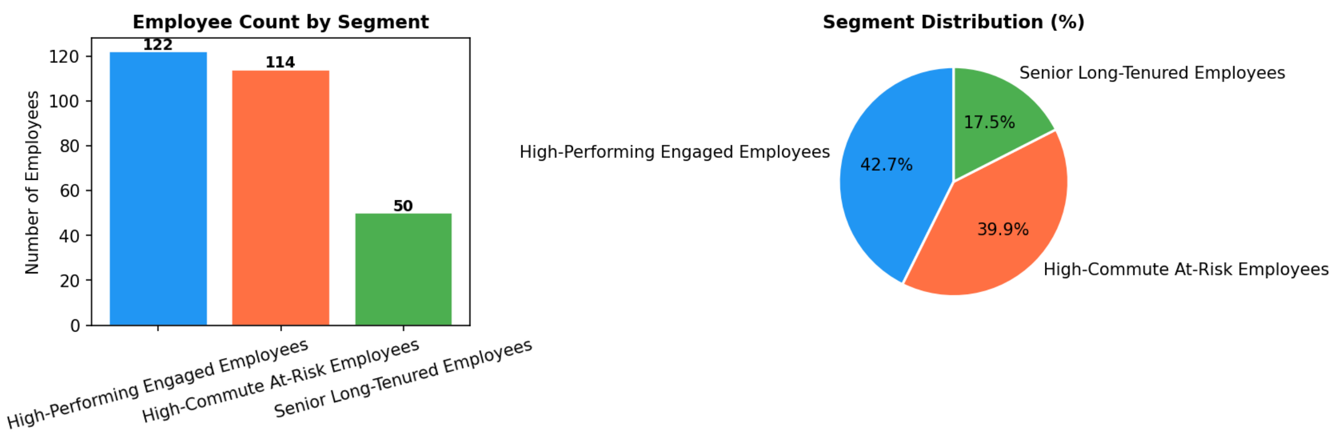

Figure 2. High-Performing Engaged Employees form the largest segment (42.7%), followed closely by High-Commute At-Risk Employees (39.9%). Senior Long-Tenured Employees are the smallest group at 17.5%, which is expected: fewer employees reach long tenure in any organisation.

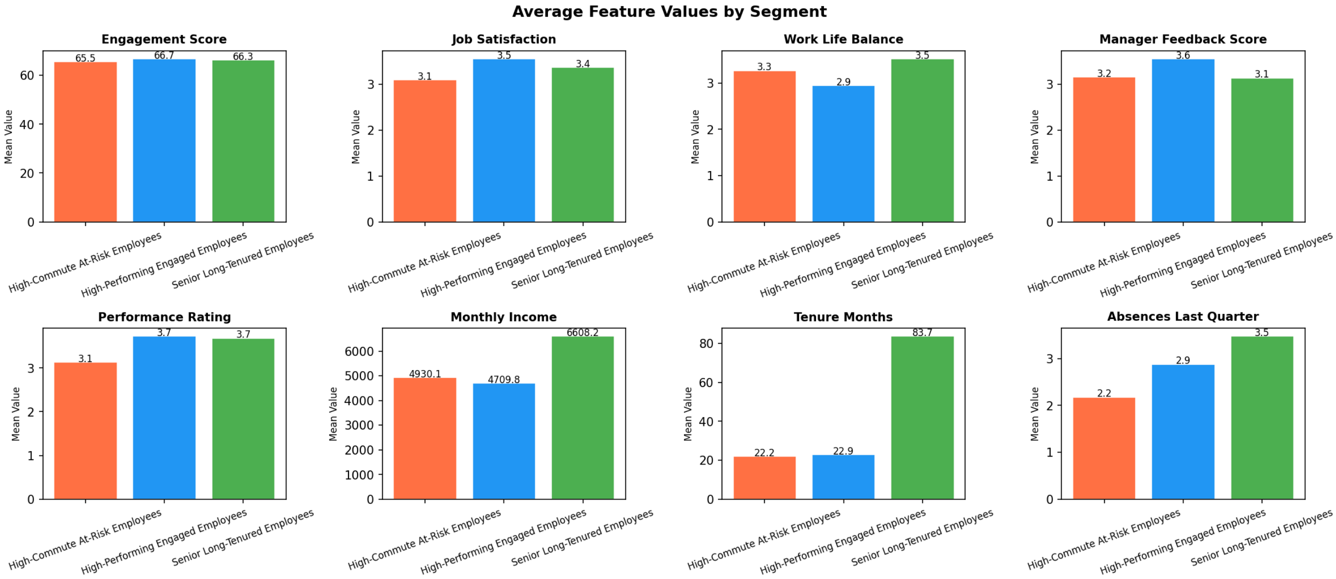

Figure 3. High-Performing Engaged Employees lead on satisfaction, performance, and manager feedback but have the lowest income despite their output. Senior Long-Tenured Employees earn the most and have the greatest tenure, but also the highest absence rate. High-Commute At-Risk Employees score lowest on engagement and satisfaction.

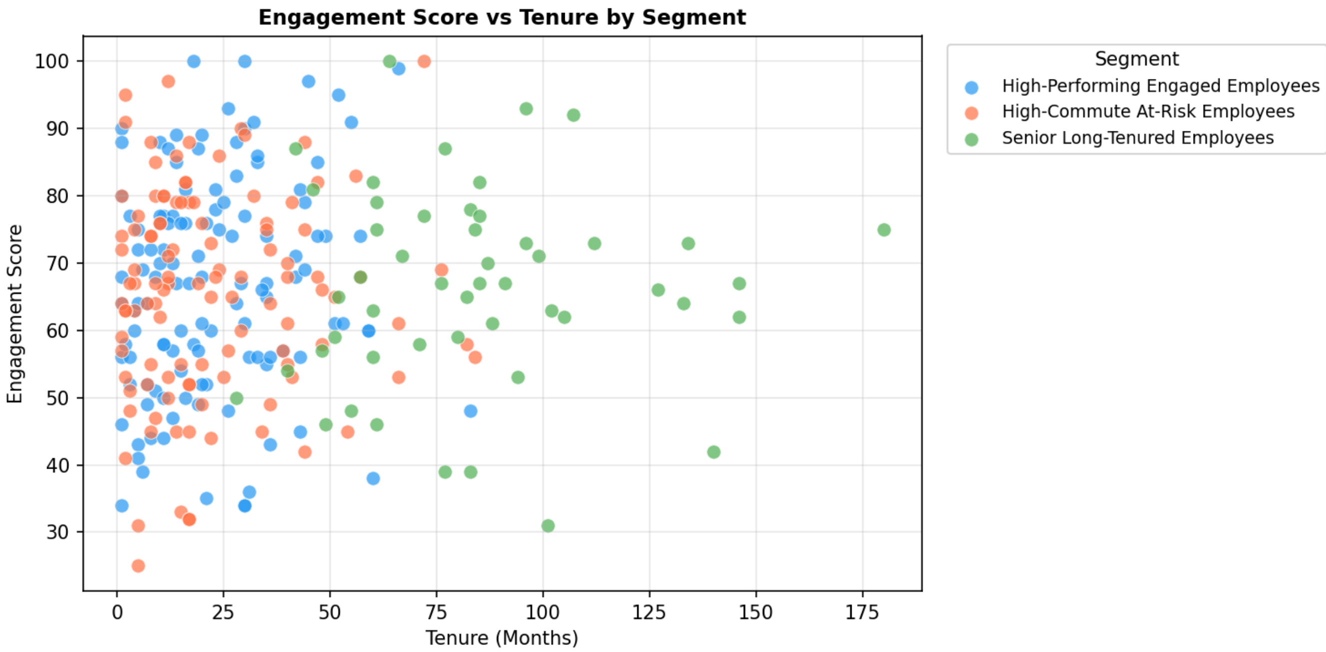

Figure 4. Green points (Senior Long-Tenured) cluster clearly to the right, reflecting high tenure. Blue and orange points overlap in the early-tenure range: engagement scores alone do not separate them. It is the commute burden that distinguishes At-Risk Employees from High Performers.

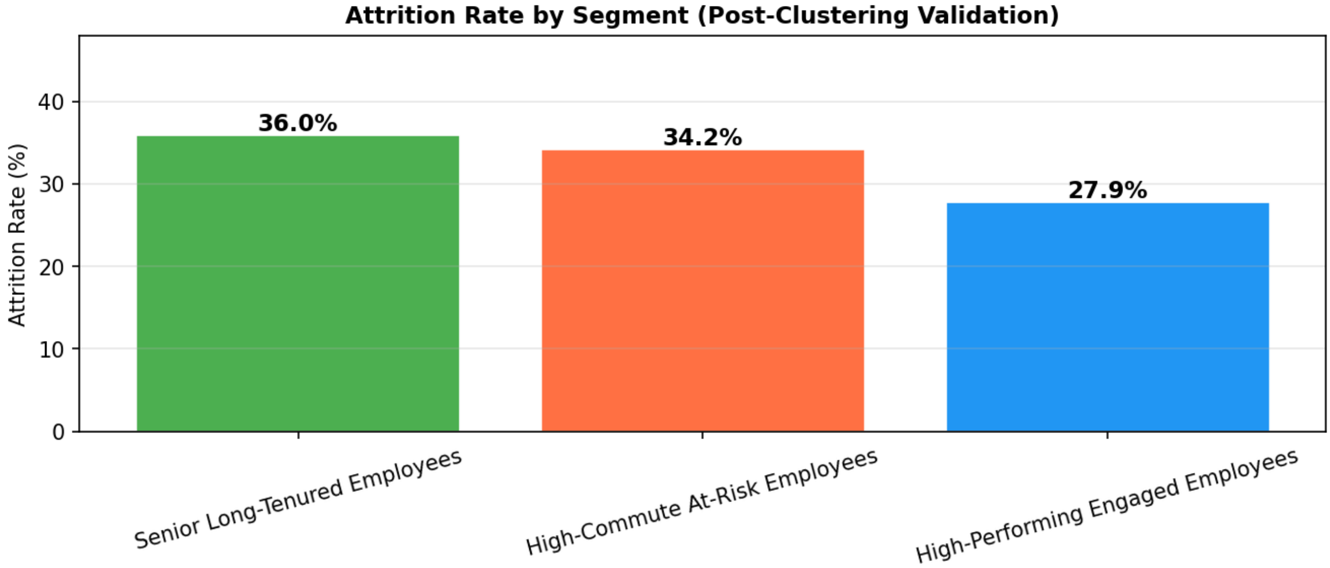

Figure 5. Post-clustering validation: Senior Long-Tenured Employees have the highest attrition rate (36.0%), followed by High-Commute At-Risk Employees (34.2%). High-Performing Engaged Employees show the lowest attrition (27.9%). Long tenure does not protect against attrition: stagnation, not financial pressure, appears to drive senior departures.

6. Business Interpretation

High-Performing Engaged Employees	122 employees, 42.7%

This is TalentWorks' most productive segment. These employees have the highest job satisfaction, performance ratings, manager feedback scores, and training hours. They live close to work and have short commutes. However, they earn the least of the three segments and have the lowest work-life balance scores, suggesting they are being stretched without proportionate compensation. Despite being the best performers, they carry a 27.9% attrition rate.

Recommended strategy: Prioritise salary reviews and compensation benchmarking. Offer flexible working arrangements to improve work-life balance. Recognise and reward high performance formally. This is the segment TalentWorks can least afford to lose.

High-Commute At-Risk Employees	114 employees, 39.9%

This is the most vulnerable day-to-day segment. These employees have the lowest engagement scores, job satisfaction, and performance ratings, combined with the longest commute times (averaging nearly 70 minutes each way) and the greatest distance from work (26.4 km). Their 34.2% attrition rate reflects the cumulative toll of commute burden on wellbeing and commitment. It is essential not to label this group as low performers without examining the structural cause: the commute is outside the employee's direct control.

Recommended strategy: Explore remote or hybrid work options. Conduct one-on-one conversations to understand individual circumstances. Consider commuter support programmes or schedule flexibility. Early intervention is critical before disengagement becomes resignation.

Senior Long-Tenured Employees	50 employees, 17.5%

This is the most unexpected finding. Despite being the most experienced and highest-paid group (average tenure of nearly 7 years, monthly income of $6,608), this segment has the highest attrition rate at 36.0%. They also have the most absences and the fewest training hours. The combination of high absence, low training investment, and high attrition suggests stagnation. These employees may feel their growth has plateaued, their contributions are no longer being developed, or that external opportunities are more attractive.

Recommended strategy: Offer mentorship roles, stretch assignments, and leadership development opportunities. Increase training investment for this group. Conduct stay interviews to understand what would make them want to remain. Losing senior employees is costly: not just in replacement cost, but in institutional knowledge.

7. Limitations

- Categorical features excluded: Department, Workload Level, Employment Type, and Remote Work Available were not included. Encoding them was possible but would have significantly expanded the feature space and reduced interpretability. Some HR-relevant structure may therefore be missing from the segments.
- Synthetic dataset: The data does not come from a real organisation. Patterns and attrition rates may not reflect real-world workforce dynamics.
- Cluster instability: K-Means is sensitive to initialisation and the choice of k. Results could differ with a different random seed or feature set.
- Static snapshot: Clustering reflects one point in time. Employee circumstances evolve, and segments should be revisited periodically rather than treated as fixed labels.
- Overlapping segments: Blue and orange points overlap considerably in the scatter plot. The boundary between High-Performing and At-Risk employees is not sharp. Some individuals may sit close to the boundary and could move between segments over time.

8. Responsible AI Reflection

Clusters are for support, not punishment. Segment labels should trigger supportive interventions such as conversations, workload review, and flexibility arrangements. They must never be used for punitive HR decisions such as performance improvement plans, reduced promotion opportunities, or redundancy consideration. Using cluster membership as grounds for negative treatment is both ethically wrong and counterproductive.

Avoid labelling employees based on structural disadvantage. Low engagement and high absences in the High-Commute At-Risk segment may reflect factors entirely outside an employee's control: long travel times, family responsibilities, or unmanageable workloads. Attributing poor outcomes to individual failure without investigating structural causes risks unfair treatment and may disproportionately affect employees who are already disadvantaged.

Features such as age and income can reflect historical bias. If the organisation has historically underpaid or underinvested in certain groups, those patterns will be embedded in the data and carried forward into the clusters. The model does not correct for this: it reflects it.

Human oversight is essential. No clustering output should automatically determine HR actions. Segment results should be reviewed by HR professionals who can apply organisational context, assess individual circumstances, and ensure interventions are fair, proportionate, and consistent with employment law and equity principles.

Transparency with employees matters. If segmentation informs HR strategy, employees have a right to know that such tools are in use and to understand how they have been categorised. Trust depends on transparency.

9. Conclusion

This midterm applied K-Means clustering (k = 3) to segment 286 TalentWorks employees into three distinct groups based on engagement, satisfaction, performance, income, tenure, and commute characteristics. The choice of k = 3 was supported by both the elbow method and silhouette score, providing quantitative justification beyond visual inspection alone.

Post-clustering validation using *Left_Company* confirmed that the segments differ meaningfully in attrition rates, validating that the groupings capture real workforce dynamics rather than arbitrary patterns. The most striking finding is that Senior Long-Tenured Employees, despite being the highest-paid and most experienced group, have the highest attrition rate (36%). This points to stagnation and disengagement, not financial pressure, as the primary driver of their departure.

Each segment requires a different type of intervention: compensation and recognition for High Performers, commute relief and flexibility for At-Risk Employees, and growth and development opportunities for Senior staff. Used responsibly as a guide for conversation and prioritisation, not as a basis for automated decisions, this segmentation can meaningfully support TalentWorks' attrition reduction goals.